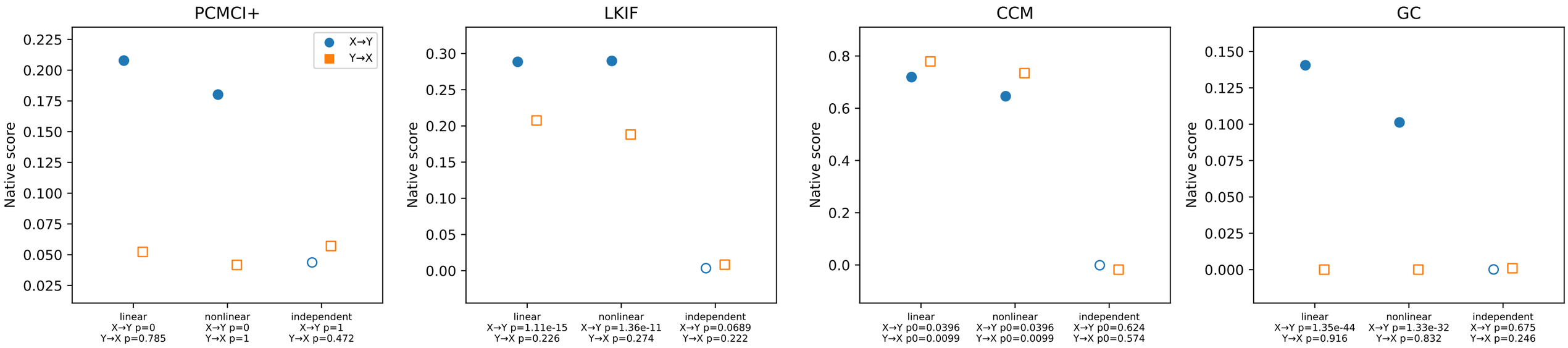
Baseline 1: original uniform process (20-yr grid)
Replicate 0; scores differ by method
PCMCI+
LKIF
CCM
GC
Native score
X→Y
Y→X
linear
X→Y p=0
Y→X p=0.785
nonlinear
X→Y p=0
Y→X p=1
independent
X→Y p=1
Y→X p=0.472
linear
X→Y p=1.11e-15
Y→X p=0.226
nonlinear
X→Y p=1.36e-11
Y→X p=0.274
independent
X→Y p=0.0689
Y→X p=0.222
linear
X→Y p0=0.0396
Y→X p0=0.0099
nonlinear
X→Y p0=0.0396
Y→X p0=0.0099
independent
X→Y p0=0.624
Y→X p0=0.574
linear
X→Y p=1.35e-44
Y→X p=0.916
nonlinear
X→Y p=1.33e-32
Y→X p=0.832
independent
X→Y p=0.675
Y→X p=0.246
Retained direction
Direction not retained
Unresolved evidence
CCM lag failed
Hollow markers include CCM lag-screen rejection. CCM score and p0 remain the original Tp=0 results.